

Course Syllabus

STAT 109: Introductory Biostatistics — Summer Session 1, 2026 (online, async)

STAT 109: Introductory Biostatistics

Cal Poly Humboldt — Summer Session 1, 2026

Course Information

Course Number: STAT 109

Course Title: Introductory Biostatistics

Units: 4

Session: Summer Session 1, 2026 (June 1 – July 2, 2026)

Mode of Instruction: Online, **asynchronous**

General Education: Lower Division GE Area B4 – Mathematical Concepts & Quantitative Reasoning

How this course works

- **Course website (home):** readings, labs, project instructions, reference sheets, and practice quiz banks.
- **Canvas:** submit work, take graded quizzes, see feedback and grades, and join module discussions.
- **Course path:** recommended order through six modules (0–5) over five weeks.

There are **no required live meetings**. Work through each module on your own schedule within its window. Modules unlock in order in Canvas ($0 \rightarrow 1 \rightarrow \dots \rightarrow 5$).

Instructor Information

Teacher: Dr. Rosanna Overholser

Email: rho3@humboldt.edu

Office Location: BSS 334

Office Phone: 707-826-4022

Office Hours: By appointment (email to schedule). I will respond to email within one business day during the session.

Please use your Cal Poly Humboldt email account to contact me and check both Canvas and your Cal Poly Humboldt email regularly for course communications.

Course Description (Catalog Description)

Descriptive statistics, probability, random variables, discrete and continuous distributions, confidence intervals, contingency tests, regression and correlation, tests of hypothesis, analysis of variance. Emphasis: methods and applications used in the biological and natural resource sciences.

Division: Lower Division · **Instruction Mode:** Online (async)

Prerequisite: MATH 101 or MATH 101I or MATH 102 or MATH 109B (may be taken concurrently with instructor approval) with a grade of C- or better or ALEKS PPL assessment score over 65 or appropriate high school coursework. See [Math Placement and ALEKS PPL](#) for details. MATH 102 may be taken concurrently with instructor approval.

Note: if you plan to use STAT 109 as a prerequisite for another MATH, DATA or STAT course, you must have a grade of C- or better in this course.

Course Learning Outcomes

By the end of this course, students will be able to:

1. Use numerical and graphical descriptive statistics to summarize and interpret data.
2. Explain the role of randomness, probability, and sampling in statistical inference.
3. Apply probability rules and work with common probability distributions.
4. Construct and interpret confidence intervals for means and proportions.
5. Conduct and interpret basic hypothesis tests.
6. Analyze relationships between pairs of variables using correlation and regression.
7. Use a statistical programming language to analyze data and communicate results.
8. Distinguish between correlation and causation and recognize limitations of statistical conclusions.

General Education & Institutional Learning Outcomes

This course fulfills **Lower Division General Education Area B4 (Mathematical Concepts & Quantitative Reasoning)** and supports Cal Poly Humboldt's Institutional Learning Outcomes related to quantitative reasoning, critical thinking, and communication.

Course Materials & Workload

This is a **five-week intensive** version of STAT 109. You should expect to spend approximately **20–25 hours per week** on readings, reference-sheet updates, labs, quizzes, discussions, and projects.

Required Materials

- **Course website:** [Course path](#), [lectures](#), [methods map](#), [R reference](#), [resources](#).
- **Software:** R in [Google Colab](#) (free; no local install required). Lab and project notebooks link to Colab from the course site. Optional: install R and RStudio on your own computer ([resources](#)).
- **Reference sheets:** You will build and update two personal references over the term — a **statistics/methods** sheet and an **R** sheet — using the course templates. You may use these on open-notes quizzes.
- **Calculator:** A basic scientific calculator is helpful for by-hand checks; most computation is in R.

Optional Materials

Textbook: [Introductory Statistics for the Life and Biomedical Sciences](#) by Julie Vu and Dave Harrington, available for free online.

Access & Affordability

Students who do not have personal access to required technology may use campus computer labs and library resources. If you want to get a free used computer, please check whether our campus's [Reusable Office Supply Exchange \(ROSE\)](#) has one available.

Fees

There are no additional course fees beyond standard tuition and registration fees.

Weekly Rhythm (Modules 1–4)

Each project module follows the same pattern on the [course path](#):

Step	When (typical)	What
Read 0 + Read 1	Early in the week	Project preview and core reading
Quiz A (Def & Notation)	After Read 1	Canvas · open notes · unlimited attempts
Read 2	Midweek	Extended reading
Quiz B (Concept Check)	After Read 2	Canvas · open notes · unlimited attempts
Demo + exercise lab	Wednesday	Course site · submit exercise lab in Canvas
Synthesis quiz	Thursday	Canvas · open notes · two attempts (highest score)
Project milestone	Friday	Short planning checkpoint in Canvas
Project Discussion	Sunday (ideal due date) Anytime in the module	Full project submission Community prompt · see below

Module 5 is four days long: lab **Wednesday July 1**; synthesis quiz, milestone, **Project 5**, and **portfolio package** due **Thursday July 2**.

Within each module window you may work ahead or slightly behind, but complete modules in order. If dates shift when the term opens, **Canvas is the official calendar**; the **course path** will be updated to match.

Course Feedback & Assessment

Component	Weight	Description
Module 0 (orientation)	3%	Start Here, tech check, introduce-yourself discussion
Quiz A + Quiz B (all modules)	10%	Low-stakes practice · open notes · unlimited attempts · lowest module pair dropped
Synthesis quizzes (5 modules)	12%	Integration · open notes · two attempts each · highest score counts
Lab checkpoints	15%	Exercise labs · completion + one rubric criterion per module
Discussions	5%	One community prompt per module · participation, not statistical correctness
Projects 1–5	50%	Applied analyses with write-ups and runnable code (slightly back-loaded toward Projects 4–5)
Portfolio package	5%	Module 5 capstone: index linking all five projects + polished summaries + brief reflection
Total	100%	See assessments for project list and details

Practice quiz banks on the course site ([quizzes/quiz-m*-*.qmd](#)) are open-ended with full answer keys. Canvas quizzes may draw a random subset and autograde where possible.

Projects

Five portfolio projects apply course methods to real questions. Each includes a one-page summary and a runnable R notebook. See [assessments](#) and individual project pages.

Project	Module	Due (2026)
Project 1 — Testing a Claim	1	Sun Jun 7
Project 2 — Estimating the Truth	2	Sun Jun 14
Project 3 — Discovering Patterns	3	Sun Jun 21
Project 4 — Designing Fair Comparisons	4	Sun Jun 28
Project 5 — Looking Ahead + portfolio package	5	Thu Jul 2

Dual deadlines (Modules 1–4): The **ideal** project due date is **Sunday**; a **grace** submission window may apply through the following day in Canvas. Module 5 has a single **Thursday July 2** deadline with no Sunday extension.

Discussions

One **community discussion** per module (Modules 0–5). These build connection in an async course; they are graded on **participation**, not statistical correctness. Prompts are on Canvas; see the course site for the belonging arc across modules.

Exams

There are **no midterm or final exams** in this async session. Exam-style items are integrated into **synthesis quizzes** and project work instead.

Grading

Final letter grades will be assigned based on overall performance:

A: 93–100% · A-: 90–92% · B+: 87–89% · B: 83–86% · B-: 80–82% · C+: 77–79% · C: 73–76% · C-: 70–72% · D: 60–69% · F: 0–59%

Note:

- **Grade Mode:** Letter grade by default. Students may elect Credit/No Credit grading according to university deadlines and procedures.
- A minimum grade of **C-** is required for the course to count toward GE Area B4.
- If you stop submitting work without communicating with me, you may receive a grade of WU (Withdrawal Unsatisfactory). Please contact me early if you are having difficulties.

Late or Missed Work Policies

- **Quizzes (A, B, Synthesis):** Complete by the due dates in Canvas. Quiz A and B allow unlimited attempts before the deadline. Synthesis allows two attempts; highest score counts.
- **Exercise labs:** Due on the module lab date (typically Wednesday). Late labs may lose credit after the due date except in documented emergencies; contact me as soon as possible.
- **Project milestones:** Short planning checkpoints (typically Friday). Submit on time so I can give feedback before the project is due.
- **Projects:** Ideal due **Sunday** with a possible **grace** window for Modules 1–4 (see Canvas). After the grace window,

late projects may receive reduced credit. **Project 5 and the portfolio package** are due **Thursday July 2** with no Sunday extension.

- **Discussions:** Post and reply by the module end date in Canvas.

Note: Technology issues (internet access, device failure, Colab outages) are not automatically valid excuses for late work; plan ahead and keep local copies of your notebooks. If Colab or Canvas is down on a due date, email me promptly with a screenshot or error message.

AI Policy

You may use AI tools (e.g. ChatGPT, NotebookLM, Copilot) to **learn** course material, explore code, or draft text — especially on take-home labs and projects.

For work you **submit for a grade**:

1. If you used AI, **acknowledge it briefly** (name of the tool is enough; you do not need to paste prompts).
2. You are **responsible** for the correctness, appropriateness, and academic integrity of everything you submit — AI-assisted or not.
3. Do not submit AI-generated analysis or text you have not reviewed and understood.

I may use AI to improve assignment wording or check clarity. I will not use AI to grade your work or run AI-detection tools on submissions.

Course Topics & Schedule

Topics (by module)

Module	Theme	Core statistical ideas
0	Start here	Course tools, R in Colab, reference sheets
1	Testing a Claim	Probability, binomial process, hypothesis test with p-value
2	Estimating the Truth	Sampling, bias, confidence intervals
3	Discovering Patterns	Descriptive statistics, graphs, tidyverse EDA
4	Designing Fair Comparisons	Study design, confounding, two-group inference
5	Looking Ahead	Correlation, simple linear regression, prediction

Full outcomes and links: [course path](#) · [lectures](#).

Important dates (Summer 2026)

Date	Event
Before Jun 1	Complete Module 0
Jun 1 – 7	Module 1 · Project 1 due Sun Jun 7
Jun 8 – 14	Module 2 · Project 2 due Sun Jun 14
Jun 15 – 21	Module 3 · Project 3 due Sun Jun 21
Jun 22 – 28	Module 4 · Project 4 due Sun Jun 28
Jun 29 – Jul 2	Module 5 · Project 5 + portfolio due Thu Jul 2

Check [Canvas](#) for exact due times. This schedule is subject to change; changes will be announced in Canvas.

University Policies & Resources

Students are responsible for reviewing the [Cal Poly Humboldt Syllabus Addendum](#), which includes policies and resources related to:

- Academic honesty
- Disability accommodations
- Emergency procedures
- Title IX
- Student conduct and support services

Accommodations

Students who wish to request disability-related accommodations should contact the Student Disability Resource Center (SDRC) as early as possible and then email me (Rosanna Overholser, rho3@humboldt.edu) your letter of accommodation.

Online Conduct

Standards of respectful and professional behavior apply to all course interactions, including Canvas discussions, assignments, and email communication.

Changes to the Syllabus

This syllabus is subject to change. Any changes will be communicated through **Canvas announcements**.
